HOPE LAB / METHOD REFERENCE

Emulsification Index (E24) Assay

A reference guide to emulsion-height measurement, calculation, controls and reporting.

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Purpose and principle

The E24 index is a simple measure of the persistence of an emulsion after a defined standing period. It is calculated from the height of the emulsion layer relative to the total liquid-column height. E24 describes emulsion formation and stability under the selected test conditions; it is not a direct measurement of surface tension or critical micelle concentration.

Method workflow (for supervisor-approved, non-viable cell-free test material)

- Define the sample matrix, test material status, selected oil phase, vessel geometry, mixing approach and reading time in the approved method before comparing samples.
- Include a matrix control without the test material and an appropriate reference emulsifier or positive comparison where available.
- Apply one consistent, validated sample-to-oil ratio and mixing procedure to all samples. This guide does not prescribe operational mixing conditions or chemical choices.
- At the specified endpoint, identify the emulsion layer using the same visual rule for every sample and record the emulsion height and total liquid-column height.
- Calculate E24 for each independent replicate. Report the oil phase, matrix, ratio, mixing method, endpoint, replicate count and any deviations.
- Repeat or confirm findings using an orthogonal method if the intended claim concerns biosurfactant identity, interfacial activity or molecular mechanism.

Calculation

E24 (%) = (Height of emulsion layer / Total height of liquid column) x 100.

Example for arithmetic only: if an emulsion layer occupies 2.0 units of a 5.0-unit total column, E24 = (2.0 / 5.0) x 100 = 40%. This is an illustrative calculation, not an experimental target or expected result.

Controls and interpretation

- Matrix-only control: identifies emulsification caused by the sample matrix itself.
- Reference emulsifier: provides a comparison when a suitable, characterized reference is available.
- Replicate measurements: report individual values and mean with SD or another justified variability measure.
- A stable visible emulsion does not by itself prove that a compound is a biosurfactant. Bioemulsifiers and low-molecular-weight surfactants can behave differently.
- Do not compare E24 values across studies unless oil phase, sample matrix, ratio, vessel geometry, mixing and endpoint are sufficiently comparable.

Interpretation limits

E24 is a screening measurement influenced by the oil phase, sample composition, vessel dimensions, mixing energy, reading time and observer-defined layer boundaries. It should be interpreted alongside complementary characterization, not as standalone chemical identification.

References and authoritative resources

1. Cooper DG, Goldenberg BG. Surface-active agents from two Bacillus species. Applied and Environmental Microbiology. 1987;53(2):224-229. doi:10.1128/aem.53.2.224-229.1987.

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